RAJIB
Che-50

2022, 2022(t), 2019(t), 2018(t), 2018, 2016, 2019

(*) Define polarization and polarizability?

⇒ Polarization is the displacement of positive charges relative to negative charges in a system. polarization is the distortion of a negatively charged ion's $e^-$ cloud by a positively charged ion. polarization occurs when an electric field distorts the negative cloud of $e^-$ around positive atomic nuclei in a direction opposite the field. $P_i = N' \alpha E_i$

Polarizability is the ability to form instantaneous dipoles. It is a property of matter. polarizability refers to the tendency of molecules to generate induced electric dipole moments when subjected to an electric field. electric field seperates the centres of the positive and negative charges. Electric field has induced a dipole moment.

$$\mu_{ind} \propto E_i$$

where, $\mu_{ind}$ is induced dipole moment and $E_i$ is strength

of electric field.

or, $\mu_{ind} = \alpha E_i$

where $\alpha$ is polarizability.

If $E_i$ = unit electric field strength, then $\mu_i = \alpha$.

2022, 2022(t), 2018(t), 2019(t), 2018, 2017

Differentiate Polar & non-polar in term of molar polarization?

⇒

| Polar | non-polar |
|---|---|
| 1. Polar molecule have permanent dipole. | 1. It Do not have parmanent dipole. |
| 2. It have orientation polarization | 2. It have no o. polarization |
| 3. Temperature dependent | 3. Independent |
| 4. It has non zero dipole moment | 4. It has zero dipole moment |
| 5. EX: HCl, $CH_3Cl$, $NH_3$, $H_2O$, $SO_2$ etc. | 5. EX: $H_2$, $Cl_2$, $CH_4$, $C_6H_6$ etc. |

2022, 2022(t), 2019(t), 2018(t), 2020, 2018, 2017, 2016

⊛ Discuss application of dipole moment ?

⇒ (i) Dipole moment helps to predict wheather a molecule is polar or non-polar.

(ii) % of ionic character can be calculated from D.P.

(iii) It helps to distinguish btwn cis- & trans- isomers.

(iv) It helps to distinguish btwn ortho, meta & para- isomers.

(v) Bond angle can' be calculated with the help of it.

(vi) It helps to determine the resonance structure.

(vii) It helps to determine the shape of molecules.

2022(t), 2021(t)

⊛ Define polar and non polar molecule ?

⇒ polar molecule is a chemical species in which the distribution of electrons btwn covalently bonded atoms is not even. Polar molecule occur when there is an electronegativity difference between the bonded atoms. EX. HCl, $H_2O$, $NH_3$, $CH_3Cl$ etc.

Non polar is a chemical species in which the distribution of $e^-$ btwn covalently bonded atoms is even. Non polar molecule occur when $e^-$ are

shared equal btwn atoms of diatomic molecule or when polar bonds in a larger molecule cancel out by each other. Ex. $H_2$, $Cl_2$, $C_6H_6$, $CH_4$ etc.

2022(t), 2015(t), 2015, 2024(t)

Explain dipole-dipole, dipole-induced dipole & induced dipole-induced dipole interactions?

⇒ Dipole-dipole interactions are attractive forces btwn the positive end of one polar molecule and the negative end of another polar molecule. It occurs btwn molecules with permanent dipoles. This is stronger than London forces but is weaker than ion-ion interaction because only partial charges are involved. H-bond is special type of D.-D. Interactions

⇒ Dipole-induced dipole interactions occurs btwn a polar molecule and a non polar molecule or atom. It is als Known as debye force. Ex. the interaction btwn HCl and Ar.

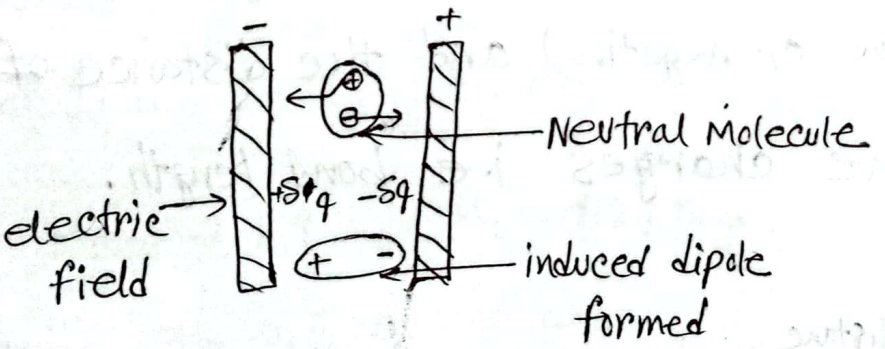
⇒ Induced dipole-Induced dipole interactions is actually Known as London force. The london force is a

temporary attractive force that results when the e in two adjacent atoms occupy positions that make the atoms from temporary dipoles. It arises due to the non-zero instant dipole moments of all atoms & molecules.

2021, 2014(1), 2018(1), 2019(1), 2018, 2017, 2016, 2019, 2029(1)

⊗ Explain distortion polarization & orientation polarization?

⇒ The creation of positive or negative charges in a neutral molecule under the influence of an electric field is known as electric polarization. It is temporary disappears as the electric field is removed. This type of polarization is known as distortion or induced polarization.




⇒ Due to thermal orientation, the molecules would occupy some mean position btwn the direction of the electric field and their original position. This effect of the field on the molecule is called orientation P..

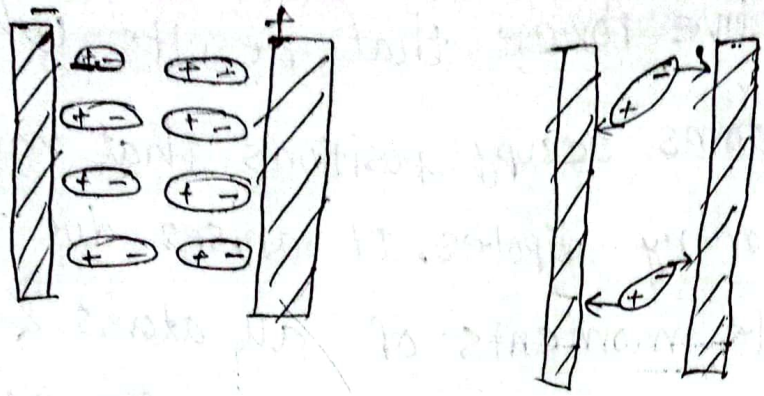


fig: orientation polarization of molecules.

(★) 2015(1), 2020, 2015, 2024 (1)
Define dipole moment ?

⇒ A dipole moment is a measurement of the separation of two opposite electrical charges. Dipole moment occurs ~~when~~ btwn two ions in an ionic bond or btwn two atoms in a covalent bond. It is vector quantity.
Dipole moment is the product of the magnitude of the charges (positive or negative) and the distance of separation btwn the charges i.e. bond length.

$$\vec{\mu} = q \cdot \vec{r}$$

dipole moment — separated charge — distance btwn charge

(★) futorial
bond length of H-I is 1.60 Å & $\mu = 0.38$ D. clclt % of ionic character of H-I bond ?

expected $\vec{\mu} = q \cdot r = 4.8 \times 10^{-10} \times 1.6 \times 10^{-8}$

$= 7.68 \times 10^{-18}$ esu-cm

$= 7.68$ D (debye)

observed $\mu = 0.38\ D$

$\therefore$ % of ionic character $= \frac{0.38}{7.68} \times 100\% = 4.95\%$

Ans.

(★) 2020 Calculate D.m., if $e^-$ is placed 1Å from equal +ve charge.

$\Rightarrow \mu = q \times r = 4.8 \times 10^{-10}\ esu \times 10^{-8}\ cm = 4.8 \times 10^{-18}\ esu\text{-}cm$

$\therefore \mu = 4.8\ D$ (debye)

(★) 2019, 2019(t) ~~Calculate~~ $\mu$ of $NH_3$ is 1.46D. If angle $\angle HNH = 108°$, calculate $\mu$ of NH bond?

$\Rightarrow \mu^2 = \mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos\alpha$ | $\alpha = 108°$, $\mu_1 = \mu_2$

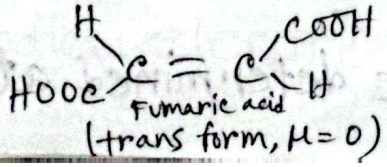
$\Rightarrow 1.46^2 = 2\mu_1^2 + 2\mu_1^2 \cos 108°$

$\Rightarrow \mu_1^2 = 1.54 \quad \therefore \mu_1 = 1.24\ D$ Ans.

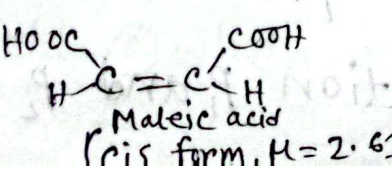
(★) 2018, 2016, 2017, 2020, 2022 Describe any one application of D.m.?

$\Rightarrow$ Distinction btwn cis- & trans- isomer:

Dipole moment help us in distinguishing btwn the cis and trans- isomers of a compound:



Fumaric acid
(trans form, $\mu = 0$)



Maleic acid
(cis form, $\mu = 2.61\ D$)

2024(t), 2021

Describe Ebert's & vapour-temp. method for M?

⇒ Ebert's Method: polar molecules in the gaseous state when subjected to an electric field will undergo both orientation & distortion polarizations. But when the same substance in the solid state is subjected to the same field, the orientation of molecules is not possible and only distortion polarization will occur. Therefore, total molar polarisation is given by,

$$P_{gas} = P_i + P_o$$

$$P_{solid} = P_i$$

∴ orientation polarization $P_o = P_{gas} - P_{solid}$

$$\frac{4}{3}\pi N \frac{\mu^2}{3KT} = P_{gas} - P_{solid}$$

$$\therefore \mu = \sqrt{\frac{9KT}{4\pi N}\left(P_{gas} - P_{solid}\right)}$$

Temp. Method: If total polarization $P_t$, we know

$$P_t = a + \frac{b}{T}$$

If total polarization $P_1$ and $P_2$ are determined at

two temperatures $T_1$ & $T_2$ respectively then,

$$P_1 = a + \frac{b}{T_1} \quad \& \quad P_2 = a + \frac{b}{T_2}$$

$$b = (P_1 - P_2)\left(\frac{T_1 T_2}{T_2 - T_1}\right)$$

The value of dipole moment, $\mu$ is easily calculated from that of $b$, i.e.

$$b = \frac{4\pi N}{9K}\mu^2$$

$$\therefore \mu = \sqrt{\frac{9K}{4\pi N} \cdot b}$$

As, $K = 1.38 \times 10^{-16}$ erg $K^{-1}$, $N = 6.023 \times 10^{23}$, $\pi = 3.1416$

$$\therefore \mu = \sqrt{\frac{9 \times 1.38 \times 10^{-16} \cdot b}{4 \times 3.1416 \times 6.023 \times 10^{23}}}$$

$$= 0.0128 \times 10^{-18}\sqrt{b}\ \text{esu-cm} = 0.0128\sqrt{b}\ \text{D}$$

Ans.

(*) 2015, 2015(t)
Explain the dependence of polarizability on frequency

⇒ (i) Debye equation is derived assuming molecules in static electric field. (ii) when applied field is

oscillating Debye eqn can not be applied when the frequency is very high. (iii) permanent dipole moment can not align with the ever changing